When Does Training on Downscaled Images Yield the Same Gradients?

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Abstract

Scale-wise diffusion schemes rest on a spectral argument: flow-matching noising progressively masks high spatial frequencies, so above some noise level a downscaled latent carries *almost* the full surviving signal. We ask whether this licenses a training-side substitution — can the gradient of a low-rank adapter be computed from a downscaled latent at high noise without changing what is learned? We give a first-order account of what the substitution can cost: the gradient gap decomposes into a noise-dependent *input term*, read through the gradient norm, plus a σ-independent *floor* carried by the network's input–output Jacobian — and we embed the spectral argument, in its strongest tolerance-parameterized form (Xiao et al., 2026), as the null model of the input term: a one-parameter family of per-route boundaries that depends on the downscale *ratio* alone, is compressed to a ~0.13 spread in σ on our measured latent spectrum, and admits no floor. Testing this takes metrology, not just measurement: naive finite-draw gap estimators *manufacture* exactly the token-count-ordered floors at issue, so we build a debiased estimator (per-arm self-floors with an attenuation correction, plus draw-count extrapolation) and state every claim at the instrument's measured resolution. The debiased measurements on a DiT-based flow-matching model reject the null where the accounts diverge: the mildest route's crossover sits at $\sigma \approx 0.5$, ~3.5× the equal-power prediction; a 2× downscale is certifiably unsafe at *every* noise level, retaining a debiased gap of ≈ 0.30 on pure-noise input; and the surviving endpoint gap equals its image-free control at every route — a floor carried by the compute graph, governed by the *absolute token count* of the downscaled grid, concentrated in early blocks, and separable into a rotary-position-embedding share and a graph residue. Debiasing also cuts both ways: about half of the intermediate route's raw floor was estimator bias; what remains (+0.056, CI $[0.043, 0.071]$) is real but leaves no certifiable safe window at our resolution. The practical output is a measured (route, σ) safety map with explicit detectability thresholds, and an opt-in σ-conditional trainer that substitutes the downscaled latent only on measured-safe steps, for a projected wall-clock ceiling of ~14% at fixed step count on our corpus (end-to-end A/B pending).

1 Introduction

The cost of a diffusion-transformer forward pass grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. That makes the title question worth asking precisely. At high noise, can the *gradient* of an adapter be computed on a downscaled latent as a drop-in substitute for the native one — same model, same objective, same σ distribution, only the spatial grid changed on some steps? On our corpus roughly half of all training draws sit at $\sigma > 0.5$, so an affirmative answer on even the mildest downscale is worth a projected ~14% of wall-clock at a fixed step budget (§5).

There is a ready-made answer, and it is the premise of a family of recent methods: under flow-matching noising $z_\sigma = (1-\sigma)x + \sigma\epsilon$, the per-frequency signal-to-noise ratio falls with σ, and high spatial frequencies drop below the noise floor first — so above a spectral crossover, a downscaled latent carries *almost* the full surviving signal. This observation underwrites scale-wise distillation (Starodubcev et al., 2025), spectral progressive diffusion (Xiao et al., 2026), pyramidal flow matching

(Jin et al., 2024), spectrally-guided noise schedules (Esteves & Makadia, 2026), and progressive-resolution training more broadly (Karras et al., 2018; Chen et al., 2024; Hoogeboom et al., 2023). By the spectral answer, substitution is safe *whenever the noise has masked the frequencies the coarse grid loses* — on our measured latent spectrum, from $\sigma \approx 0.14$.

The measured answer is different in structure, not just in degree. Safety is a per-route boundary jointly governed by two quantities the spectral answer cannot express together: the downscale *ratio* governs a noise-dependent data term, and the *absolute token count* of the target grid governs a σ-independent floor carried by the compute graph itself. The mildest route's measured boundary sits at $\sigma \approx 0.5$ — 3.5× the equal-power prediction, and no tolerance in the spectral family reproduces it — while a 2× downscale is certifiably unsafe at *every* noise level, retaining a gap of ≈0.30 on pure-noise input (§4.3, Fig. 1). The evidence collected for the methods above is collected where they operate — at the level of generated output — and a small quality-score difference does not transfer to a small difference in what a training step *learns*.

Answering the question honestly is, unexpectedly, a metrology problem. The natural gap estimator — a redraw floor minus a single demoted-arm cosine — is biased by finite-draw variance, and the bias grows as the demoted grid shrinks: it *manufactures* exactly the token-count-ordered floor signature under test. Every claim in this paper is therefore stated in debiased units, at the instrument's measured resolution.

The contributions:

1. **Gap metrology: a debiased estimator and a certification resolution** (§3.1, §4.2). We introduce per-arm self-floors with a split-half attenuation correction and a draw-count extrapolation $\mathrm{gap}(D) = \mathrm{gap}_\infty + c/D$ (an uncorrected native self-cosine reads 0.85 at the noise endpoint; corrected, it extrapolates to 1.005, CI [0.994, 1.016]), and derive the instrument's median certification resolution $\varepsilon^*(N, D)$: "safe" is a one-sided non-inferiority statement at that resolution, never an absolute claim.

2. **A theory of the demotion gap, stated before it is fit** (§3). From the perturbation expansion and cosine geometry we derive, unconditionally, a four-term angular expansion of the gap — data share, graph share, projected interaction, remainder — and, under four named assumptions, its reduction to a two-term account: a route amplitude on a universal mismatch shape read through the U-shaped gradient norm, plus a σ-independent graph floor. The spectral argument, in its strongest tolerance-parameterized form (SPD, Xiao et al., 2026), enters as the null model of the data term's input-mediated part — ratio-only, and floorless by construction.

3. **Debiased measurements that discharge the theory's predictions and reject the null where the accounts diverge** (§4). The floor exists, is σ-flat, equals its image-free control at every route, and decomposes into a rotary coordinate-system share and a graph residue; ratio governs the amplitude while absolute size governs the floor; the account *predicts* held-out routes at ∼0.07–0.09 RMSE, with the floor law hitting two floors it never saw. The exact angular form of the account resolves a systematic overshoot its quadratic reduction leaves (§4.7).

4. **The answer, in deployable form** (§5). A measured (route, σ) safety map at stated resolution — {1024→896 at $\sigma > 0.5$; 1280→1024 at $\sigma \gtrsim 0.75$}, no certifiable window for 1024→768, certifiably unsafe everywhere for →512 — and an opt-in σ-conditional trainer that substitutes the downscaled latent only on measured-safe steps, for a projected wall-clock ceiling of ∼14% at fixed steps (end-to-end A/B pending).

The presentation is theory-first; the practice was observational-first: verdict criteria were frozen before each probe ran, the account was pre-registered with an outcome matrix (including a kill switch) before the discriminating runs, and the debiasing campaign's decision rules — including the rule under which our own headline floors would have been retracted as variance artifacts — were pre-registered before it ran. Each claim's epistemic status (pre-registered / confirmatory / post-hoc) is consolidated in the claims ledger (Table 4) rather than hedged sentence-by-sentence.

2 Related work

Flow matching and DiT. We work in the flow-matching / rectified-flow formulation (Lipman et al., 2023; Liu et al., 2023; Esser et al., 2024) on a Diffusion Transformer (Peebles & Xie, 2023) with rotary position embeddings (Su et al., 2024). Adapters are LoRA (Hu et al., 2022).

Scale-wise and progressive-resolution diffusion. Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogeboom et al., 2023) train at increasing resolution as a schedule. A more recent family ties resolution to the noise level itself, on a spectral premise. Under flow-matching noising, each spatial-frequency band of the signal retains an effective signal-to-noise ratio that falls as σ grows, and the finest bands sink below the noise floor first; the noise level above which a band is masked is that band's *crossover* σ. Above the crossover of the bands a coarse grid cannot represent, a downscaled latent should therefore carry almost the full surviving signal. Pyramidal flow matching (Jin et al., 2024) builds its training and sampling pyramid on this premise, and scale-wise distillation (SwD) (Starodubcev et al., 2025) states the claim directly: above the crossover, the noised downscaled latent is close in distribution to the downscaled noised latent. Spectral Progressive Diffusion (SPD) (Xiao et al., 2026) gives the argument its sharpest available form: a tolerance-parameterized per-frequency activation time under a diagonal Gaussian spectral model — a criterion on the *Bayes-optimal predictor's* per-band output, not merely on input SNR — from which an inference-time resolution schedule follows. SPD also fine-tunes (with LoRA) against *stage-specific modified velocity targets* along the progressive trajectory — changing the objective per resolution stage rather than substituting the latent under the native objective, which is our question; that they observe a training–inference gap motivating this fine-tuning is independently consistent with our finding that resolution changes are not free at the training level. Spectrally-guided noise schedules (Esteves & Makadia, 2026) apply the same per-frequency reasoning to schedule design, deriving bounds on the useful minimum and maximum noise levels from the image spectrum — the spectral account operating as the field's working model of what noise masks. We are explicit about scope: neither SPD nor SwD *claims* naive gradient equivalence under latent substitution — SwD validates its pipeline at the output level, and SPD fine-tunes precisely because resolution changes are not free. What we test is the tempting extension of their shared premise — that spectral sufficiency of the input licenses a training-side swap — not their methods. Our contribution is orthogonal to all of these: we measure whether input-level closeness transfers to the *training gradient*, holding model and objective fixed — and find it does not transfer at any tolerance.

Positional extension. Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* — matching the downscaled grid's relative phase geometry to native — to decompose the resolution-sensitivity floor, and adopt the σ-gated band schedule of Zhao et al. (2026) as a training-time refinement.

3 Theory: the demotion gap and a two-term account

3.1 Estimand and resolution

Objective and noising. We first define the training objective whose gradients are under test. A flow-matching model is trained to predict the constant-velocity transport between a clean latent and pure noise. With clean latent x, noise $\epsilon \sim \mathcal{N}(0, I)$, and a noise level $\sigma \in (0, 1]$, the noised input is the linear interpolation $z_\sigma = (1-\sigma)\,x + \sigma\,\epsilon$ — at $\sigma \to 0$ the input is the clean latent, at $\sigma = 1$ it is pure noise — and the regression target is the velocity $v = \epsilon - x$ that carries z_σ along the interpolation. With caption condition c, the loss is $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$, and gradients are taken with respect to the LoRA adapter parameters only; the backbone is frozen.

Demotion. A *demotion* $e_{\text{nat}} \to e$ replaces the native latent with a coarser-grid one, produced exactly as a scale-wise trainer would produce it. With y the source image, $\mathcal{R}_e$ the pixel-space downscale to the tier-e bucket (native aspect ratio preserved), and $\mathcal{E}$ the VAE encoder,

$$x_{\text{nat}} \;=\; \mathcal{E}\big(\mathcal{R}_{e_{\text{nat}}}(y)\big) \quad\longrightarrow\quad x_e \;=\; \mathcal{E}\big(\mathcal{R}_e(y)\big),$$

a latent with proportionally fewer spatial tokens; we never downsample in latent space. Everything else about the training step is held fixed: the caption c, the σ under test, and the noise draw structure (the same stratified draw scheme, with ϵ drawn at the tier-e shape). The demoted step then trains on $z_\sigma = (1-\sigma)\, x_e + \sigma\, \epsilon$ with target $v = \epsilon - x_e$, so the substitution changes both what the network sees and what it is asked to predict — but only through the spatial grid; nothing else in the step references the resolution.

The estimand. Three *arms* recur throughout: nat, the native arm, which trains on x_{nat}; e, the demoted arm, which trains on x_e; and reenc, a control arm that decodes and re-encodes the native latent at native resolution — it pays the VAE round trip that demotion necessarily pays, with no grid change. For an arm $a \in \{\text{nat}, \text{reenc}, e\}$, let $\bar{g}_a(\sigma) = \mathbb{E}_\epsilon[\nabla_\theta \mathcal{L}]$ be the population per-bin mean adapter gradient of a query (image, caption). The *population demotion distance* is

$$d_e(\sigma) \;=\; 1 - \cos\big(\bar{g}_{\text{nat}}(\sigma),\, \bar{g}_e(\sigma)\big) \;=\; \tfrac{1}{2}\left\|\hat{g}_{\text{nat}} - \hat{g}_e\right\|^2, \tag{1}$$

with $\hat{g}$ the unit-normalized gradients; $d_e \geq 0$ always. The *reported* quantity of every bin-resolved map and every fit in this paper is the *re-encoding excess* $\Delta_e(\sigma) = d_e(\sigma) - d_{\text{reenc}}(\sigma)$ — the demotion distance beyond what the control already pays; Δ_e can be negative, d_e cannot. Endpoint and floor tables report debiased estimates of d_e with the control as its own row; each table states which object it reports.

In a batched training scenario the same definition admits two aggregations over a probe set, and the aggregation operator is part of the estimand: per-example (cosine per image, then mean — the batch-1 worst case) or batch-aggregate (gradients summed over the batch before the cosine — the object SGD actually follows). These are different estimands, and no coefficient fitted on one is guaranteed to transfer to the other; the application section publishes one map for each (§5.1).

Resolution: what an instrument can certify. Any estimate of Eq. 1 compares finite-draw estimates of population directions — the general situation of gradient-noise-scale analysis (McCandlish et al., 2018) — so every claim carries the instrument's resolution. For a route whose true excess is zero, the paired estimate $\bar{\Delta}$ fluctuates with standard error $\text{SE}(N, D, \sigma\text{-bin})$; considering its one-sided 95% upper bound (1.645 SE for a zero-mean estimate), we define the *certification resolution*

$$\varepsilon^*(N, D, \text{bin}) \;=\; 1.645\, \text{SE}(N, D, \text{bin}), \tag{2}$$

the smallest margin certifiable even for a truly safe route, and "safe at margin ε" as the non-inferiority statement

$$\text{UB}_{95}\big(\bar{\Delta}\big) \;<\; \varepsilon, \qquad \varepsilon \geq \varepsilon^*. \tag{3}$$

A "safe" verdict is thus a statistical non-inferiority claim at instrument resolution, never a hard bound, and a true gap below ε^* is undetectable by construction; throughout, "no certifiable window" means a route's upper bounds never cross below ε at the current (N, D), while "certifiably unsafe" means its lower bound stays above it.

3.2 Data and graph perturbations

We now derive the perturbation demotion applies to the gradient. Differentiating the loss of §3.1 in the adapter parameters gives, per draw and up to a constant,

$$g \;=\; \nabla_\theta \tfrac{1}{2}\left\|\hat{v}_\theta - v\right\|^2 \;=\; J^\top r,$$

with $r = \hat{v}_\theta - v$ the prediction residual and $J = \partial \hat{v}_\theta / \partial \theta$ the Jacobian through which the compute graph turns a residual into an adapter gradient. Demotion changes exactly the two factors of this product — the residual, because it changes the data the objective consumes, and the Jacobian, because it changes the compute graph the gradient flows through — and nothing else about the map from (image, noise, caption) to adapter gradient. Averaging over draws and expanding the product of the perturbed factors — an elementary but bookkeeping-heavy step, written out in Appendix A together with what exactly the remainder collects — the perturbation $\delta g = \bar{g}_e - \bar{g}_{\text{nat}}$ decomposes as

$$\delta g \;=\; \underbrace{J^\top \Delta\bar{r}(\sigma)}_{\text{data branch } B_e} \;+\; \underbrace{\Delta J^\top\, \bar{r}}_{\text{graph branch } C_e} \;+\; \underbrace{\Delta J^\top \Delta\bar{r} + \cdots}_{\text{remainder } R_e}, \tag{4}$$

where:

1. **the data branch** B_e: demotion changes the *data* the objective consumes — spatial-frequency bands above the coarse grid's Nyquist limit are absent from the input *and* from the target $v = \epsilon - x$ (and the re-encode perturbs what survives) — a perturbation $\Delta\bar{r}(\sigma)$ of the mean prediction residual. Its input-mediated part shrinks as the model's reliance on input detail shrinks with σ; its target-mediated part does not: perturbing the image by δx perturbs the residual by $D_z f\,[(1-\sigma)\,\delta x] + \delta x$, and the second term carries unit weight at every σ. At σ=1 the input is pure noise on every grid, so the input-mediated part vanishes in distribution — but the endpoint is *not* data-free by construction; whether the surviving target-mediated share is resolvable is an empirical question, discharged in §4.4.
2. **the graph branch** C_e: token count, rotary phase density, sequence-length-dependent normalization — an operator perturbation ΔJ that is σ*-independent*, because the grid change does not reference σ.

One qualification is worth stating precisely: J_{nat} and J_e act on different grids, so the two branches are not literal matrix products across arms; they are defined *operationally*, as the two independent perturbation directions in the shared adapter-parameter space where both gradients live, isolated by the designated probes of §4.4.

This decomposition already shows why input-level spectral sufficiency cannot bound the gradient. Input noise masking alone provides no bound on expected-gradient disagreement, because systematic target and graph terms survive noise averaging: the quantity training consumes is $\mathbb{E}_\epsilon[\nabla_\theta \mathcal{L}]$, the target carries the clean image at coefficient one at every σ, and ΔJ never references the spectrum at all. Input-level sufficiency licenses inference-time substitution; it does not extend to training-gradient equivalence.

3.3 Angular geometry: the exact form and its expansions

It remains to read the perturbation of Eq. 4 in the units of the estimand, Eq. 1 — a pure exercise in cosine geometry, whose forms below are what every measurement in §4 is fit against (step-by-step derivation in Appendix B). With $G(\sigma) = \|\bar{g}_{\text{nat}}\|$ and $P_\perp$ the projector orthogonal to $\bar{g}_{\text{nat}}$, split the perturbation into a rescaling and a rotation: $\kappa_\parallel = \hat{g}_{\text{nat}}^\top \delta g \,/\, G$ and $\kappa_\perp = \|P_\perp \delta g\| \,/\, G$. Substituting $\bar{g}_e = \bar{g}_{\text{nat}} + \delta g$ into Eq. 1 then gives, *exactly*,

$$d_e \;=\; 1 - \frac{1}{\sqrt{1+\kappa_{\text{eff}}^2}}\,, \qquad \kappa_{\text{eff}} \;=\; \frac{\kappa_\perp}{1+\kappa_\parallel}\;: \tag{5}$$

the gap is a saturating function of a single rotation-to-scale ratio — only the orthogonal component rotates the gradient direction, and a parallel component merely renormalizes it. Taylor-expanding Eq. 5 at small κ ($d_e = \frac{1}{2}\|P_\perp \delta g\|^2/G^2 + \cdots$) and substituting the branch decomposition $\delta g = B_e + C_e + R_e$ of Eq. 4 into the square gives the *four-term angular expansion* — like Eq. 5, derived unconditionally:

$$d_e(\sigma) \;\approx\; \underbrace{\frac{\|P_\perp B_e\|^2}{2G^2}}_{\text{data share } S_e} \;+\; \underbrace{\frac{\|P_\perp C_e\|^2}{2G^2}}_{\text{graph share } F_e} \;+\; \underbrace{\frac{\langle P_\perp B_e,\, P_\perp C_e\rangle}{G^2}}_{\text{projected interaction } I_e} \;+ R'_e, \tag{6}$$

where R'_e now has an exact characterization (Appendix B): it collects the remainder branch R_e, the beyond-quadratic terms of Eq. 5, and the parallel-component correction. S_e and F_e are non-negative; I_e can carry either sign — a fact that matters in §4.7.

The reduced two-term account. Under four named assumptions — **(A1)** small remainder: R'_e negligible over the claimed domain; **(A2)** graph-relative stationarity: $\|P_\perp C_e\|/G$ is σ-constant (numerator and denominator share the residual's scale), so $F_e = \text{Floor}_e$; **(A3)** data-branch factorization: $\|P_\perp B_e\| \approx a_e\, m(\sigma)$ — a σ-independent route amplitude times a route-independent mismatch shape; **(A4)** negligible projected interaction: $|I_e| \ll S_e + F_e$ over the claimed domain — the expansion reduces to the *two-term account*

$$\text{gap}_e(\sigma) \;\approx\; \underbrace{A_e \cdot \big(m(\sigma)/G(\sigma)\big)^2}_{\text{data term}} \;+\; \underbrace{\text{Floor}_e}_{\text{graph term}}\,, \qquad A_e = a_e^2/2, \tag{7}$$

where $m(\sigma) = \|\Delta\bar{r}(\sigma)\|$ is the mismatch of the model's mean prediction residual across grids (directly measurable, §4.5) and $G(\sigma)$ is the total gradient norm. The quadratic power on m/G is fixed by the cosine geometry *locally*; far from the small-mismatch regime the licensed read is the parent form itself: keeping Eq. 5 un-expanded, with A3's loading written as $\kappa_e(\sigma) = c_e\, m(\sigma)/G(\sigma)$ (so $A_e = c_e^2/2$ in the small-κ limit), yields a data term that *saturates* instead of overshooting when κ is comparable to one. The loading itself — that the orthogonal gradient mismatch is proportional to the residual mismatch, $\|P_\perp \delta g\| = c_e\, m(\sigma)$ — is an *empirical* ingredient, not a derivation; so wherever a fitted functional form is reported, the un-expanded Eq. 5 is labeled the *best empirical predictor* and Eq. 7 the *derived small-perturbation form*; §4.7 scores both, held out.

The graph term has identifiable parts. Changing the grid changes (i) the rotary coordinate system — every band of the position embedding accumulates phase at a different density across the image — and (ii) non-positional graph statistics: attention softmax over a different token count, sequence-length-dependent normalization, and the sheer capacity of the coarse graph to approximate the fine graph's computation. Splitting the graph share $\mathrm{Floor}_e = \mathrm{RoPE}_e + \mathrm{Resid}_e$ yields one sharp prediction: the positional share is a *coordinate-system artifact*, so an exact positional-interpolation intervention — evaluating the demoted grid at fractional rotary coordinates that reproduce the native relative phase geometry — should remove RoPE_e wherever the data branch is inactive ($\sigma \to 1$); and a band-counting sketch says the residue should grow with absolute grid change (the number of rotary bands whose per-extent rotation count crosses decorrelation grows with the extent), attaching an absolute-size governor to Resid_e. We flag the sketch's epistemic status plainly: like YaRN's own band thresholds (Peng et al., 2024), its constants are calibrated, not derived; we claim no closed form for Floor_e.

The ledger. Explicitly: the exact form (Eq. 5) and the four-term expansion (Eq. 6) are derived unconditionally from Eqs. 1 and 4 (Appendix B); the two-term reduction (Eq. 7) is derived *conditional on A1–A4*, each of which is tested by a designated probe (§4.1); the coefficients A_e (or c_e), $m(\sigma)$, $G(\sigma)$, Floor_e are measured, not derived; and any fitted power or loading is labeled empirical. The account was pre-registered with an outcome matrix (including a kill switch: all endpoint gaps ≈ 0 would have falsified the graph term) before the discriminating runs.

3.4 The spectral account as a null model of the data term

The spectral argument, made precise, is a model of one part of one branch — and its strongest available form is SPD's tolerance-parameterized crossover (Xiao et al., 2026). Under the modeling assumption $x_0^{(\omega)} \sim \mathcal{N}(0, P_\omega)$ per frequency ω (with P the clean-latent power spectrum), their Proposition 1 states that for all $\sigma \geq t_\omega$ the *Bayes-optimal* per-band velocity prediction is within tolerance δ of pure noise, $\mathbb{E}[|v^{*(\omega)} - \epsilon^{(\omega)}|^2] \leq \delta$, and their Proposition 2 places the safe-substitution boundary for a coarse grid at the t_ω of its Nyquist band ω_e. In the terms of the angular expansion (Eq. 6), the null is therefore the claim that the gap is the data share alone, gated at the Nyquist band's activation time — no graph share, no interaction:

$$\mathrm{gap}_e^{\mathrm{null}}(\sigma) \;=\; S_e(\sigma)\, \mathbf{1}\big[\sigma < t_{\omega_e}\big], \qquad t_\omega \;=\; \frac{1}{1 + \sqrt{\delta \,/\, \big(P_\omega\, (1 + P_\omega - \delta)\big)}}, \qquad F_e = I_e = 0. \quad (8)$$

This is a prediction-level criterion, strictly stronger than raw input SNR; the often-quoted equal-power crossover $\sigma_{\mathrm{eq}}(f) = \sqrt{P(f)}/(1 + \sqrt{P(f)})$ is its $\delta = (1 + P_\omega)/2$ slice.

Its scope must be stated exactly: the null constrains only the *input-mediated part of the data branch* B_e. It does not constrain the target-mediated share, the graph branch C_e, the interaction I_e, or the Jacobian gain through which a residual mismatch becomes a gradient mismatch. Read as the null model of the data term, Eq. 8 has three structural properties, each checkable before any gradient is measured:

(N1) One-parameter family. All route boundaries move together under the single tolerance δ; the family is totally ordered.

(N2) Ratio-only. In normalized frequency the demoted grid's Nyquist sits at $f_{\mathrm{cut}} = \frac{1}{2}\, e/e_{\mathrm{nat}}$ — a function of the downscale *ratio* alone. The null therefore cannot distinguish routes of equal ratio at different absolute sizes, and assigns nearly identical boundaries to nearly equal ratios regardless of grid size. It is structurally blind to any absolute-size effect.

(N3) No floor. For every route and every δ, the null's gap vanishes above a finite boundary: every route becomes safe at some σ.

And under the same diagonal-Gaussian assumption the null extends to the residual level: the posterior mean is a per-band Wiener shrinkage — under a Gaussian prior and Gaussian noise the minimum-mean-square-error estimator is linear with per-band gain $P_\omega/(P_\omega + n_\omega)$, the classical Wiener filter (Wiener, 1949) — so with no cross-frequency coupling the cross-grid mean-residual mismatch is confined to the destroyed band and **(N4)** ordered across routes by destroyed-band energy.

The family is instantiated on our measured latent spectrum in §4.3 (Table 2); its curve-level read in gap units necessarily borrows our bridge — the null emits residual units only, so any gap-unit prediction is *the null transported through the account's G and a calibrated route gain*, not a parameter-free prediction (§4.7). Two properties of the instantiation are fixed before any gradient is measured: the latents are high-frequency-quiet, so the family is *compressed* — the spread between the mildest and harshest route never exceeds ~0.13 in σ at any tolerance — and the two routes our measurements will separate maximally (1024→896, ratio 0.875, and 896→768, ratio 0.857) receive boundaries within 0.01 of each other at *every* δ.

3.5 Finite draws and the debiased estimator

The estimand of Eq. 1 is defined on population mean gradients; any instrument estimates them from D noise draws. This finite-draw step is not a neutral nuisance: in principle it biases the naive estimate in exactly the direction under test, so the correction belongs to the theory, not to the lab notebook.

Why finite draws manufacture floors. The natural estimate of Eq. 1 — a redraw floor minus a single demoted-arm cosine — is asymmetric: the floor is a cosine between *two* finite-draw estimates of the native gradient, while each demote arm contributes *one* estimate compared against the native mean. Finite-draw noise attenuates every such cosine below its population value (each estimated direction carries variance the population direction lacks), and the attenuation is arm-dependent: per-draw gradient variance grows as the token count falls, because the MSE loss averages over fewer elements, so a demoted arm's single estimate is noisier than the native estimates it is compared to. An iso-direction null — demotion changing *nothing* but the variance — therefore still produces a positive gap, and one that *grows as the target grid shrinks*: the same signature as "absolute token count sets the floor." A token-count-ordered floor can thus be manufactured entirely by the estimator, and no naive read can tell the two apart — which is why every verdict in this paper is stated in debiased units.

The debiased estimator. Two corrections, used together. First, *self-floors with attenuation correction*: every arm runs a second independent draw set g'_d, giving a per-arm self-cosine $\cos_{\text{self}} = \cos(g_d, g'_d)$ — each arm gets the same two-estimate structure the native floor has. The debiased cosine is the classical split-half disattenuation correction,

$$\hat{c} \;=\; \frac{\cos(\bar{g}_{\text{nat}},\, \bar{g}_d)}{\sqrt{\cos_{\text{floor}} \cdot \cos_{\text{self},d}}}, \qquad \widehat{\text{gap}} \;=\; 1 - \hat{c}, \tag{9}$$

which cancels the finite-draw attenuation of numerator and denominator to first order. Second, *draw-count extrapolation*: the attenuation of an accumulated-draw cosine decays as $1/D$, so sweeping nested draw subsets and fitting $\text{gap}(D) = \text{gap}_\infty + c/D$ reads off the draw-limit gap gap_∞ directly. The two corrections are mutually checking: if the disattenuation removes the finite-draw component, debiased estimates must come out D-flat ($c \approx 0$) under the sweep — a prediction the instrument verifies (§4.2).

4 Experiments

4.1 Discriminating predictions and designated probes

The account and the null agree that gaps shrink with σ and order by severity. Everything else is discriminating. Each prediction below names the designated probe that discharges it in the subsections that follow; several double as falsifiers of the account's own assumptions (A1–A4).

1. **High-σ persistence** (null: every gap vanishes as $\sigma \to 1$; account: persistence equal to Floor_e, surviving pure-noise input) — endpoint probe, §4.4. Kill switch: all endpoint gaps ≈ 0 falsifies the graph term.
2. **Endpoint ≡ x-zero ≡ target-α-flat** (the endpoint's surviving target-mediated share is below resolution — the empirical discharge that the endpoint reads the graph term) — x-zero probe and target-strength sweep, §4.4. This also tests A2's premise that the floor is a graph property.
3. **Governors** (null: ratio-only, N2; account: ratio governs A_e while absolute target size governs Floor_e) — iso-severity twin and cross-tier transfer, §4.5.
4. **Residual route-uniformity** (null: route-ordered residual mismatch, N4; account's A3: a route-independent shape $m(\sigma)$) — residual probe, §4.5.
5. **Curve shape** (null: gap shape tied to per-band SNR; account: m/G read — an interior peak where the U-shaped G is smallest, low-G amplification, and a low-σ dip that renormalizing by G removes) — §4.5.
6. **Held-out predictivity** (the reduced account, fit on three routes, must predict routes it never saw from its governors alone) — held-out validation and functional-form refit, §4.7.
7. **The reduction's domain** (where A2/A4 fail, the reduced form may err in a signed way: a negative I_e can pull the measured gap *below* the additive floor, which no positive two-term form can do) — the designated direct probe is a demote–re-promote arm isolating the data branch with Floor = 0 by construction (**[pending]**, §4.7).

The experiments discharge these one term at a time: the instrument and its validation (§4.2), the phenomenon and the null's boundary-level score (§4.3), the graph floor (§4.4), the data branch and its governors (§4.5), the floor's decomposition (§4.6), and the account confronted with routes it never saw (§4.7).

4.2 The instrument, validated

Model and data. All measurements use Anima (CircleStone Labs & Comfy Org, 2025), an open DiT-based flow-matching text-to-image model (2B parameters, 28 transformer blocks, 3D rotary position embeddings, and the Qwen-Image VAE, Wu et al., 2025), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length $e \in \{512, 768, 896, 1024, 1280\}$; a tier-$e$ image occupies roughly $(e/16)^2$ latent-patch tokens (e.g. ~4,100 tokens at 1024, ~3,000 at 896, ~2,160 at 768, ~1,000 at 512). The probe adapter is a plain LoRA checkpoint trained at native tiers; we return to this operating-point dependence in §6.

The gradient probe. For each image and each of B uniform σ-bins we accumulate adapter gradients over D stratified noise draws per arm (verdict runs: 8×8 unless noted; $N = 40$ images) and compare arms by cosine similarity of the flattened accumulated gradients:

- **floor**: two independent draw sets at native resolution — the redraw null;
- **reenc**: decode–re-encode at native resolution — the control of §3.1;
- **demote arms**: one per edge e under test;
- **self-floors** (debiased runs): the per-arm second independent draw set the debiased estimator of §3.5 requires.

The gap is in cosine units (a mismatch *fraction*, scale-free; bin-mean SEM ~0.02 at N=40). Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with $\|g\|$ across bins. Every bin-mean curve must pass a split-half reliability check over images — a discipline forced on us by an earlier failure in which per-image rankings from the same gradients had split-half reliability indistinguishable from zero. Uniform σ-bins make the training-time σ-density irrelevant to the map; density-weighting the bins reproduces our earlier pooled measurements as a consistency check. Estimator details and the endpoint / x-zero probe modes are in Appendix C.

Finite-draw bias, measured. The bias mechanism of §3.5 is live at our operating point, and large. A variance back-of-envelope predicts spurious endpoint gaps of $\approx 0.02/0.05/0.15$ for 896/768/512 — plausibly ~40% of the uncorrected intermediate floors — and a retroactive scan confirms the confound: the 1024→896 endpoint gap decays as +0.100/+0.035/−0.016 at $D = 4/8/16$, a clean c/D decay with fitted $c \approx 0.5$. The native floor itself makes the point sharpest: at D=8–16 the uncorrected endpoint self-cosine reads 0.79–0.85, but a nested draw sweep ($D = 4 \dots 64$) extrapolates it to 1.005 (bootstrap 95% CI [0.994, 1.016]) — the population-level per-bin native gradient is a *fixed direction*, and the entire redraw floor is finite-draw noise, not intrinsic gradient stochasticity. No conclusion in this paper survives on the naive estimator alone.

The debiased estimator, operated. Endpoint-mode runs sweep nested draw prefixes $D \in \{4, 8, 16, 32, 64\}$ (the D=64 pass contains the smaller draw sets — one set of forwards) and fit $\text{gap}(D) = \text{gap}_\infty + c/D$ per route with a bootstrap CI over images; verdict-grid runs compute Eq. 9 per image and bin. The mutual check of §3.5 passes: debiased fits are D-flat ($|c| \leq 0.05$ for the 512 route, against $c \approx +0.29$ uncorrected) — the attenuation correction removes exactly the $1/D$ component the sweep isolates. One caveat delimits where Eq. 9 is readable. At D=8 per bin, the *unpaired* debiased estimator is unstable in bins where the floors themselves are small (mid-σ; the denominator is noisy): there the readable object is the *paired per-image excess* $\Delta_i = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$ of §3.1, with non-finite and $|\Delta| > 1.5$ values trimmed — pairing cancels the shared per-image, per-bin draw structure. All verdict maps below report this paired object. Finite-draw cosines are also compiler-kernel-path sensitive, so all pairings live within one run (Appendix C).

Measured resolution. At the verdict grid's operating point (N=40, D=8/bin) the bin-level ε^* (Eq. 2) is 0.03–0.08; endpoint-mode draw sweeps (D up to 64) tighten it to $\approx$0.01–0.02.

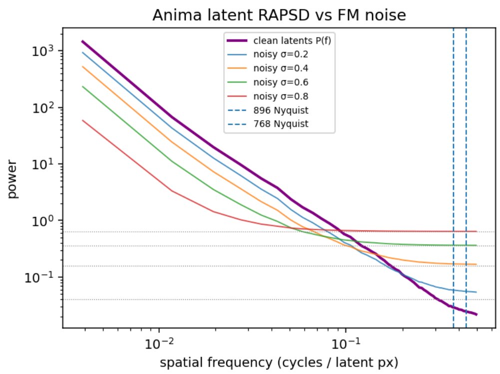
Units and coverage, stated once. From here on, every number in the main text is debiased; the uncorrected historical tables are in Appendix E and are not cited as evidence. Debiasing is not a formality — it moved verdicts in both directions: it *raised* the mid-σ gaps the attenuation had hidden, halved the 768 endpoint floor, and surfaced a small real 896 endpoint gap that single-estimate variance had buried (revision record: Appendix Table 7). Four secondary probes (the 1280-tier iso-severity twin, the 896-tier transfer, the depth split, and the positional-interpolation intervention) predate the debiased instrument; their reads are retained only where one of two defenses applies — the finite-draw bias is strictly positive, so an in-band read is conservative; and paired same-grid contrasts cancel the bias to first order — and their tables are likewise in Appendix E.

4.3 The phenomenon: gap curves against the spectral family

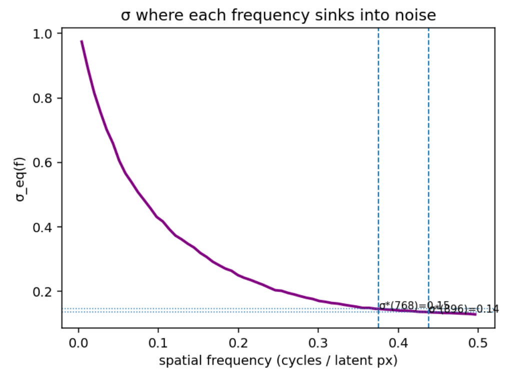
The spectrum. Anima's VAE latents are high-frequency-quiet ($P(f) < 1$ above $f \approx 0.16$ cycles/latent-pixel; latent variance 0.42), giving the equal-power crossovers $\sigma^*(896) = 0.136$, $\sigma^*(768) = 0.146$, $\sigma^*(512) \approx 0.20$, with tight per-image spread (p10–p90 $\approx$ 0.11–0.16): the spectral crossover is image-generic, and at that tolerance demotion would be safe over almost the entire σ axis (Fig. 1, top row).

The measurement. Table 1 gives the debiased verdict map for 1024-tier natives demoted to 896/768/512 (the uncorrected historical curves are Appendix Table 6).

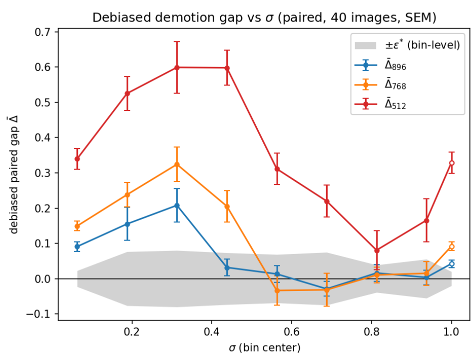
Three reads, against criteria frozen before the runs: (i) σ-dependence is real and in the predicted direction, and tier ordering holds at every bin — the *qualitative* spectral intuition survives; (ii) the crossover for the mildest route is $\sigma \approx 0.5$ (the 896 route is clearly gapped through $\sigma \leq 0.31$, marginal at 0.44, and its means sit within ±0.03 of zero from 0.56 on) — 3.5× the equal-power prediction, and no tolerance rescues the family: the δ that matches this boundary ($\delta = 0.025$, Table 2) predicts



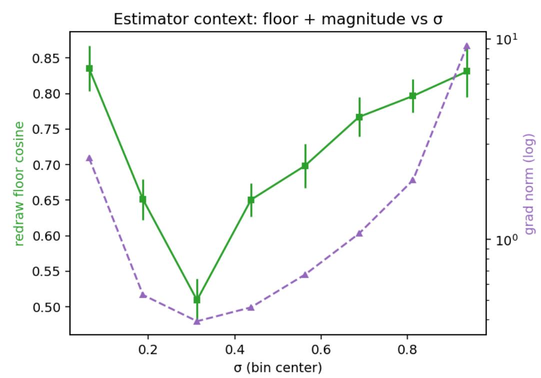
(a) Latent RAPSD against the flow-matching noise floor at four noise levels; dashed lines mark the demoted grids' Nyquist cuts.



(b) The equal-power slice $\sigma_{\mathrm{eq}}(f)$ of the spectral family. Predicted σ^*: 0.136 / 0.146 / ~0.20 for demotion to 896 / 768 / 512.



(c) Debiased demotion gap per σ-bin, paired against the re-encoding control (1024-tier natives, N=40; values in Table 1). Gray band: bin-level $\pm\varepsilon^*$ (Eq. 2); open markers: the σ=1 endpoint bin. The 896 route enters the band at $\sigma \approx 0.5$; 512 never does.



(d) Estimator context: the redraw-floor cosine and the U-shaped gradient norm $G(\sigma)$ through which every gap is read (§4.5).

Figure 1: The spectral account (top row: computed from the measured RAPSD, predicting safety from $\sigma \approx 0.14$) versus the gradient measurement (bottom row: safe only from $\sigma \approx 0.5$, and only on the mildest route). No tolerance reproduces the measured pattern (Table 2): the equal-power crossover is off by ~3.5× and the 2× downscale never becomes safe.

Table 1: **Debiased verdict map** (plotted in Fig. 1c): paired per-image debiased excess $\bar{\Delta}$ (SEM) against the re-encoding control, 1024-tier natives (N=40, D=8/bin plus a σ=1 endpoint bin at the same draws; non-finite and $|\Delta| > 1.5$ per-image values trimmed, retaining 35–40 images per cell). Bin-level ε^* at this (N, D) is 0.03–0.08 (Eq. 2).

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94	1.0
896	.091	.155	.208	.032	.013	−.029	.016	.004	.043
(SEM)	.013	.047	.047	.024	.024	.021	.024	.021	.011
768	.149	.238	.324	.206	−.034	−.032	.011	.015	.092
(SEM)	.014	.034	.049	.045	.042	.047	.022	.034	.012
512	.340	.525	.599	.598	.312	.220	.081	.166	.329
(SEM)	.029	.048	.073	.050	.045	.046	.056	.061	.030

768 safe at 0.52 and 512 safe at 0.62, while SPD's own published $\delta = 0.01$ predicts all three routes safe by 0.72; (iii) the 2× downscale route is *certifiably unsafe at every noise level*: its paired debiased lower bound stays above ε^* in every bin (means +0.08 to +0.60), including $+0.17 \pm 0.06$ at $\sigma = 0.94$,

Table 2: The spectral null (Eq. 8) instantiated on the measured latent spectrum: predicted safe boundaries t^* per route as the tolerance δ sweeps its range. SPD's published default is $\delta = 0.01$; the equal-power slice is the classical crossover. No row reproduces the measured pattern (last line; §5.1): the δ that matches the safe route's boundary predicts both failing routes safe by $\sigma = 0.63$, and no δ can produce a floor.

tolerance δ	1024→896	1024→768	1024→512	
$(1 + P_\omega)/2$ (equal power)	0.14	0.15	0.20	classical crossover
0.05	0.41	0.43	0.53	
0.025	0.50	0.52	0.62	tuned to the measured 896 gate
0.01	0.61	0.63	0.72	SPD's published default
0.005	0.69	0.71	0.79	
measured, debiased	≈ 0.5	none certified	unsafe, all σ	floors .02–.04 / .06–.09 / $\approx$ .3

where the input is ~97% noise by power in every band the coarse grid cannot represent — the high-σ persistence of prediction 1 in §4.1, which no member of the null family can produce. There is no noise level at which a 2×-downscaled latent yields the native gradient.

Two revisions against the historical record are part of the result. The 768 route's high-σ plateau was roughly half estimator bias: debiased means sit within noise of zero across $\sigma \in [0.56, 0.94]$, but with upper bounds that never cross below the 0.02 margin at this (N, D), and a real endpoint floor of +0.092 ± 0.012 remains — so "never safe" softens to *no certifiable window at instrument resolution* (§5.1). Conversely the 896 route's endpoint shows a small *real* gap (+0.042 ± 0.011) that single-estimate variance had buried; its safe window is the interior $\sigma \in (0.5, 0.94]$, not the endpoint itself.

Ratio does not transfer — the null's blind spot, measured. Re-running the probe one tier down (896-tier natives, demote to 768/512) fails its pre-registered bar: 896→768 (ratio 0.857, nearly identical to the passing 0.875) retains a high-σ residual roughly 2× the 1024→896 plateau and outside the re-encoding band (Appendix Table 9), and the debiased floors of the *same target grids* reproduce the separation independently — +0.02–0.04 on the 896-token-count grid versus +0.06–0.09 on the 768 grid (§4.4). This is exactly property N2's blindness made concrete: the null assigns these two routes boundaries within 0.01 of each other at every tolerance, and the measurement separates them into *safe* versus *no certifiable window*. "One tier down at high σ" is not the invariant; §4.5 locates why (the floor is governed by absolute size, and the 768-grid floor alone fills the 0.02 margin).

The tolerance family, scored at the boundary level. Table 2 instantiates Eq. 8 on the measured latent spectrum (P at the 896/768/512 cuts = 0.025/0.029/0.065). Because the latents are high-frequency-quiet, P barely varies across the cuts and the family is compressed: the spread between the mildest and harshest route never exceeds ~0.13 in σ at any tolerance. No row reproduces the measured pattern.

Which tolerance? Eq. 8 needs a δ, and it is worth asking where one could legitimately come from. SPD's published $\delta = 0.01$ is an inference-side choice, selected on a speed–quality Pareto ablation over generated images (Xiao et al., 2026) — nothing ties it to gradients. Machine precision is not the relevant floor either: bf16 rounding on unit-scale per-band amplitudes injects power of order 10^{-5} or below, orders of magnitude beneath any boundary-moving tolerance. Our setting, however, fixes a tolerance *by construction*: the demotion recipe pays a resize–VAE-encode round trip whose latent error has measurable per-band power $D(f)$ — the same pipeline noise the re-encoding control prices at the gradient level — and D is expressed in exactly Eq. 8's units (per-band power against unit noise). Band content below $D(f_{\mathrm{cut}})$ is invisible to the substitution pipeline even with no resolution change, so $\delta_{\mathrm{reenc}}(e) := D(f_{\mathrm{cut}}(e))$ anchors the null with zero free parameters. On the probe set this floor and its implied boundaries are **[pending: `reenc_noise_floor.py`, queued]**; we will report the anchored row alongside the family in Table 2. Either way the confrontation does not hinge on the choice: the structural failures (N2, N3) are δ-independent.

Table 3: The floor, debiased. **Endpoint**: at σ=1 the input is exactly ϵ on every grid, so the input-mediated data term vanishes by construction. **x-zero**: the image is removed from input *and* target; any surviving gap is pure graph-shape sensitivity. Debiased draw-limit values $\widehat{\text{gap}}_\infty$ [bootstrap 95% CI over images] from nested draw sweeps (endpoint: N=12, $D = 4 \dots 64$; x-zero: N=40, $D = 4 \dots 32$). The endpoint and x-zero columns agree within CI at every route, and the account's pre-registered kill switch (all endpoint gaps ≈ 0) does not fire.

route	endpoint $\widehat{\text{gap}}_\infty$ [95% CI]	x-zero $\widehat{\text{gap}}_\infty$ [95% CI]
native self-floor (cosine)	1.005 [0.994, 1.016]	—
re-encoding control	−0.003 [−0.017, +0.008]	—
1024→896	+0.019 [+0.010, +0.030]	+0.034 [+0.017, +0.058]
1024→768	+0.056 [+0.043, +0.071]	+0.074 [+0.053, +0.094]
1024→512	+0.304 [+0.197, +0.424]	+0.283 [+0.232, +0.332]

4.4 The floor is graph: three probes, one number

Three designated probes (predictions 1–2 of §4.1) establish the graph term — and with them the premises of the reduction. At the $\sigma = 1$ *endpoint* the input is pure ϵ — identical in distribution across arms — so the input-mediated data term is zero by construction; per §3.2, the target-mediated share need *not* vanish there, so the endpoint alone does not isolate the graph term — the next two probes discharge that step empirically. The debiased endpoint floors are +0.019/+0.056/+0.304 for 896/768/512 (Table 3); the 512 value clears the pre-registered confirmation bar ($\widehat{\text{gap}}_\infty \geq 0.15$) on 12 of 12 probe images individually — the token-count floor is not an estimator artifact.

The *x-zero* probe removes the image from input *and* target everywhere, isolating pure graph-shape sensitivity — and its debiased draw-limit gaps are statistically equal to the endpoint gaps at *every* route (Table 3). The *target-strength sweep* closes the remaining loophole from the α side: re-running the endpoint with target $\epsilon - \alpha x$, $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ (N=12, D=16, one run, shared draws across α), the paired debiased excess is α-flat at the well-conditioned anchors (768: +0.070 ± 0.015 at α=0 versus +0.049 ± 0.010 at α=1; 512: +0.269 ± 0.041 versus +0.337 ± 0.067; re-encoding control slope +0.003) — no resolvable target-content share. The interior points $\alpha \in [0.5, 0.75]$ are unreadable by construction and excluded: there the native residual $\alpha x - \hat{x}$ passes near cancellation, the gradient norm dips ($60 \rightarrow 33$) and the redraw floor falls to 0.87, so the estimator reads through a small G — the low-G amplification of prediction 5, surfacing along the α axis. Endpoint ≡ x-zero ≡ α-flat: three independent probes converge on one number per route. The endpoint gap *is* the graph floor, as an empirical result — established at our operating point, not assumed by construction.

How does the floor relate to the interior high-σ bins? For 512 the debiased window means (+0.08–0.60) sit at or above the floor, as additivity requires. For 896 and 768 the floors (0.02–0.04 and 0.06–0.09) are *at or below* the interior bins' certification resolution ($\varepsilon^* = 0.03$–0.08 at N=40, D=8): the endpoint sweeps resolve them only because their effective draw budget is an order of magnitude larger. The claim "the high-σ plateau is the floor" therefore sharpens to: the floor is what the plateau converges to when the data term dies and the estimator is given enough draws to see it.

Two further exonerations. The re-encoding control sits at zero within CI — debiased −0.003 [−0.017, +0.008] — so the VAE round trip is not the cause (and, the converse asymmetry: the round trip has *measurable* input-level error, yet its gradient cost is below instrument resolution — input-level error neither implies nor excuses gradient-level cost). And a forward-only probe of the model's caption-conditioned *prior* across grids shows route distances that are flat where the floors are strongly ordered — so the floor's route-ordering lives in the Jacobian factor J, not in the model's resolution-conditioned prior, and not in a training-distribution discontinuity (the never-trained 1280 grid is no more distant from 1024 than heavily-trained adjacent pairs are from each other).

4.5 The data branch: a route-uniform numerator over a U-shaped denominator

$m(\sigma)$ is route-uniform — the residual-level null fails. Measuring the mean prediction residual $\bar{r} = \mathbb{E}_\epsilon[\hat{v} - (\epsilon - x)]$ per grid and σ — the exact r in $g = J^\top r$ — shows a strong, monotone σ-shape (cross-grid excess $0.89 \rightarrow 0.36$ in relative L_2 from σ=0.125 to 1) that is the *same for every route* within ±0.02, across routes whose gradient floors span 0 to 0.3 and whose crossovers span 0.5 to nonexistent (Appendix Table 10). This kills the diagonal null at the residual level (N4): with no cross-frequency coupling, its mismatch is confined to the destroyed band and ordered by destroyed-band energy, which differs substantially across these routes — route-uniformity says the real mismatch is carried by cross-band structure the diagonal model cannot express. All route identity therefore lives in J, not in the prediction error — a pre-registered null that would equally have falsified the account's A3 had the residual curves been route-ordered; instead it *establishes* A3's shape factorization directly. The decay is smooth, Wiener-like posterior shrinkage with no hard gate at the spectral crossover: gap curves read as "collapsed" where the data term sinks below the instrument's certification resolution, not at σ_{eq}.

$G(\sigma)$ shapes the curve. Because the gap is a cosine it measures a mismatch *fraction*; the total gradient norm is U-shaped in σ ($2.6 \to 0.4$ at $\sigma \approx 0.3 \to 9.3$; Fig. 1d) — large at high σ where the model is pulling composition out of the prior and the residual is image-scale, small in the mid-σ refinement regime, and inflated again at low σ by irreducible ϵ. A monotone absolute mismatch divided by a U-shaped norm yields exactly the measured interior-peaked gap curves: bin-mean gap tracks $1/\|g\|$ (Spearman +0.55 to +0.90 across arms and runs, and within images), and renormalizing by G flips the anomalous low-σ dip into the monotone maximum in every arm of both runs — prediction 5, discharged. This is a post-hoc consistency analysis (so labeled in the ledger): it explains the *shape*; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

The governors: ratio sets the amplitude, absolute size sets the floor. With the floor isolated, what orders the routes? Two probes give a clean division of labor (prediction 3). *Absolute size, not ratio, sets the floor*: the route 1280→1024 is *more* aggressive by ratio (0.80) than 896→768 (0.857, no certifiable window), yet its floor is ≈ 0 and it enters the re-encoding band at $\sigma \gtrsim 0.75$ — refuting any pure ratio rule for the floor, and with it the null's only degree of freedom. Conversely the floor grows monotonically as the *target* grid shrinks (0.02–0.04 / 0.06–0.09 / ≈ 0.30 at target ~3,000/2,160/1,000 tokens), consistent with "floor = how well the coarse graph approximates the fine graph's computation." *Ratio, not absolute size, sets the amplitude*: a synthetic iso-severity route 1280→1120 — edge ratio 0.875 exactly matched to 1024→896, but 1.6× its absolute target size — reproduces the 1024→896 curve bin-for-bin and its crossover $\sigma^* \approx 0.5$, while the same-native 1280→1024 (ratio 0.80) separates and floors later (Appendix Table 8). Amplitude follows ratio; with $\mathrm{Floor}_{1120} \approx 0$ this completes the division: *ratio sets A_e; absolute target capacity sets* Floor_e. The pre-registered contrast (ratio-governed $\Rightarrow \sigma^* \approx 0.5$; capacity-governed $\Rightarrow$ earlier floor) discriminated cleanly. Note what survives of the spectral account here: ratio *is* the right governor for the data term — the null's error is not its governor but its scope, mistaking one term for the whole gap.

4.6 Decomposing the floor: depth, and a positional share

Depth, not module type. Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) shows the floor concentrated in *depth*: early blocks (0–9, peaking at 3–8) carry ~3× the late-block gap, uniformly across every module type within a block (at $512/\sigma{=}1$, every type sits in 0.22–0.31; Appendix Table 11). The mechanism picture: the first ~10 blocks build grid-calibrated token statistics; the divergence propagates into every parameter type's gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Notably, query and key projections show *zero* excess over value projections — which we initially read as refuting a rotary-embedding mechanism. That inference was wrong, in an instructive way.

$\mathrm{Floor}_e = \mathrm{RoPE}_e + \mathrm{Resid}_e$. Landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. The discriminating experiment is the origin-side intervention of §3.2: evaluate the demoted grid with positionally-interpolated fractional rotary coordinates (patch i at $i \cdot H_{\mathrm{nat}}/H_{\mathrm{dem}}$ per axis), making its relative phase geometry exactly match native. At the $\sigma{=}1$ endpoint this *erases* the 768 floor (paired $\Delta = +0.081$, 78% of images improved) and removes ~30% of the 512 floor (paired $\Delta = +0.096$), while leaving the in-band 896 control untouched (Appendix Table 12; the paired Δs are same-grid contrasts at matched draws, in which the finite-draw bias cancels to first order). Re-anchored against the debiased floors: the +0.081 erasure accounts for essentially all of the 768 floor ($\mathrm{Resid}_{768} \approx 0$);

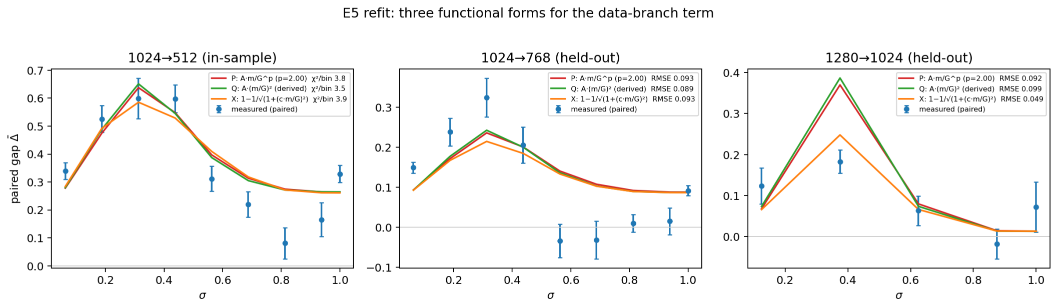


Figure 2: The account confronted with held-out routes, under three functional forms for the data term (shared floor structure): the fitted-power form, the derived small-perturbation quadratic (Eq. 7), and the exact angular link (Eq. 5). Left: the severe 512 fit route. Middle/right: the held-out routes 1024→768 and 1280→1024 — every curve is predicted from governors fitted on *other* routes. The exact angular link's saturation removes the quadratic's peak overshoot on 1280→1024; the 768 mid-σ dip below the predicted floor is form-independent (see text).

against the 512 floor of ≈ 0.30, the $+0.096$ erasure leaves a $\mathrm{Resid}_{512} \approx 0.2$ bulk. The mild-route floor is mostly a *coordinate-system artifact*; the harsh-route bulk is a genuine non-positional graph residue (attention-over-N statistics, sequence-length-dependent normalization, capacity). The capacity governor of §4.5 attaches to Resid_e, as the band-counting sketch anticipated. The 896 route's small floor (0.02–0.04) sits below the PI probe's paired resolution and remains undecomposed.

The floor as a ledger. Combining the designated probes, the floor decomposes additively, each share measured by its own intervention:

$$\mathrm{Floor}_e = \underbrace{\mathrm{reenc}}_{\approx 0 \text{ by control}} + \underbrace{\text{target content}}_{\approx 0 \text{ by x-zero / } \alpha\text{-flat}} + \underbrace{\mathrm{RoPE}_e}_{\text{erased by PI at } \sigma=1} + \underbrace{\mathrm{Resid}_e}_{\text{remainder}}$$

with totals 0.02–0.04 (896; undecomposed), ≈ 0.07 (768; RoPE the large majority, Resid $\approx$ 0), and ≈ 0.30 (512; RoPE $\approx$ 0.10, Resid $\approx$ 0.20). One caveat on the first share: the re-encoding control (native decode→re-encode) is a *proxy* for the pipeline cost demotion actually pays (source downscale→encode); the demote–re-promote arm of prediction 7 closes that gap empirically and is **[pending]**.

The positional share is a noise-regime artifact only. A σ-resolved follow-up closes the tempting route this opens (training 1024→768 through interpolated coordinates): with image content in the input, the stretched forward is *off-manifold* — the interpolated arm is *worse* than the plain demoted arm through $\sigma \in [0.56, 0.81]$ (paired -0.05 to -0.11) and wins only at $\sigma \geq 0.94$. The floor's positional share is erasable only where the input is noise-dominated, so the decomposition is a mechanism finding, not a training lever — and the data term of the 768 route is fatal on its own (+0.09–0.22 over control across the window). A frequency-*selective* banded alignment does survive in-window (§5.2).

4.7 The account confronted: held-out routes and the reduction's domain

The predictions so far test the account's *structure*. Prediction 6 tests whether it is a predictive theory: fit the reduced account on three routes, then predict two routes it never saw, from governors alone. Fit routes: {1024→896, 1024→512, 1280→1120}; held-out: {1024→768, 1280→1024}. Per fit route the excess curve $\bar{\Delta}_e(\sigma)$ is fit with a route amplitude and floor on the measured route-uniform $m(\sigma)$ and each run's own $G(\sigma)$; two governor models — $A(\text{ratio})$ interpolating the fitted amplitudes, and an exponential floor law $F(n) = F_0\, e^{-n/\tau}$ in target tokens n — then generate the held-out predictions with zero per-route freedom. Pass criteria were pre-registered in the analysis script before the numbers.

Held-out results. All four pre-registered gates fire: (i) held-out 1024→768: RMSE 0.093 — *better than an oracle quadratic fit on the held-out data itself* (0.105), and beating the null on its committed region ($\sigma > \sigma_{\mathrm{eq}}$, where the null predicts gap 0: RMSE 0.164 null vs 0.096 ours); (ii) held-out

1280→1024: RMSE 0.092 vs oracle 0.056 — inside the pre-registered 2× gate; (iii) the ratio governor holds at the amplitude level: the two ratio-0.875 routes agree at $z = 0.14$ despite a 1.6× difference in target capacity; (iv) the floor law $F(n) = 0.70\, e^{-n/1041\,\text{tok}}$, fit on the 512 and 896 floors, predicts $F(4825) = +0.007$ for the 1120 grid (fitted +0.002) and $F(2160) = +0.088$ for the held-out 768 grid — against a measured endpoint floor of +0.092 ± 0.012, dead-on. A post-hoc leave-896-out addendum (so labeled) re-derives the governors from 512+1120 alone and predicts the 896 floor at +0.019 against the measured +0.019 [+0.010, +0.030]: the floor law is 2-for-2 on floors it never saw.

Functional form: the exact link headlines. Refitting the same pipeline under three forms for the data term — a fitted power $A\, m/G^p$ with p shared and free, the derived quadratic $A\,(m/G)^2$, and the exact angular link (Eq. 5) — resolves the form question in the geometry's favor twice over (Fig. 2). First, the free exponent's own optimum lands on $p = 2.00$ exactly: the "empirical" power form recovers the derived quadratic on its own. Second, the exact link is the best held-out predictor (mean RMSE 0.071 vs 0.093–0.094 for the other two), and its entire margin comes from the 1280→1024 route (RMSE 0.049 vs ≈ 0.095): the quadratic systematically *overshoots* that route's mid-σ peak, and the exact link's saturation removes the overshoot — what a first analysis had booked as convexity in the $A(\text{ratio})$ governor was quadratic-form error. The governors are form-invariant (floor law $F(2160) = +0.087$–0.088 under all three; ratio-twin $z \leq 0.44$). Per the ledger of §3.3: Eq. 5 is the best empirical predictor (empirical through its linear loading $c_e\, m$), and the derived quadratic is its small-κ limit — one geometry at two orders, not two competing models.

Stated limits. The prediction is *not* within certification resolution anywhere (χ^2/bin 7.6–9.2 held-out under the quadratic): the licensed claim is shape, magnitude class, and governors at ~0.07–0.09 RMSE — never "predicts within ε^*." And the $A(\text{ratio})$ interpolation rests on two distinct ratio values.

The reduction's domain: one signed failure, and its signature. The 768 route's mid-σ window is the reduction's one structural failure — and its shape is diagnostic. Measured excess sits at zero across $\sigma \in [0.56, 0.94]$, *below* the predicted $\text{Floor}_{768} \approx 0.09$: a positive two-term form cannot produce a gap under its own floor, under any coefficients. In the four-term expansion (Eq. 6) this is exactly the signature of a negative projected interaction $I_{768}(\sigma) < 0$ in that window — equivalently, graph-relative stationarity (A2) fails there: the reduced account fails precisely where its named assumptions say it can, and the derived four-term form represents the failure rather than excusing it. The window therefore *delimits the reduction's domain*; the demote–re-promote probe of prediction 7 — which isolates the data branch with Floor = 0 by construction and reads I_e per bin as $\Delta_{\text{demote}} - \Delta_{\text{repromote}} - \text{Floor}_e$ — is the designated direct test, with the pre-registrable sign $I_{768} < 0$ in-window **[pending]**.

The null, scored. Against the discriminating predictions of §4.1, the null loses each one it is party to: the boundary pattern is {≈ 0.5, no certifiable window, certifiably unsafe everywhere} (§4.3); gaps persist at pure-noise input, equal to the floor and to the image-free control (§4.4); ratio governs the amplitude while absolute size governs the floor (§4.5); the residual curves are route-uniform under route-ordered gaps (§4.5); and the curve shape follows the m/G read, not per-band SNR. The account also retrodicts the observations that motivated no probe: (i) $\sigma^* \approx 0.5$ rather than 0.14 — the crossover is where the data term sinks below certification resolution, a property of the ratio, so the 3.5× discrepancy is expected, not anomalous; (ii) the 896→768 failure (the 768-grid floor alone fills the certifiable margin — no σ-gate can rescue it); (iii) 1280→1024 flooring *later* ($\sigma^* \approx 0.75$) but cleanly (harsher ratio ⇒ larger A_e; large target ⇒ Floor ≈ 0) — the single observation that breaks both pure-ratio and pure-capacity route-ordering stories. A curve-level overlay — each published tolerance δ converted to a predicted gap curve through the account's bridge (destroyed-band Bayes-residual mismatch, measured G, route gain calibrated on the one safe route) — is **[pending: analysis over existing rows; the anchored line additionally waits on `reenc_noise_floor.py`]**; unit honesty requires saying it will be *the null read through our bridge*, since the null emits residual units only.

A corollary worth stating plainly: the safety of 1024→896 is an *empirical smoothness statement about this network's function across nearby token counts*, not an information-theoretic statement about the data. There was never a reason to expect a universal ratio invariant.

The claims ledger. Table 4 consolidates every claim's epistemic status — the paper's hedging, in one place.

Table 4: Claims ledger: every load-bearing claim, its epistemic status, its designated probe, and where it is discharged. *Pre-registered* = criteria frozen before the run; *confirmatory* = designated probe run under frozen instrumentation, criteria stated with the probe; *post-hoc* = analysis after seeing the data, so labeled.

claim	status	probe	where
Token-count floor exists, carried by the graph	pre-registered (incl. kill switch)	endpoint sweep	§4.4
Endpoint ≡ x-zero ≡ α-flat (no target share)	confirmatory	x-zero, α-sweep	§4.4
896 crossover $\sigma \approx 0.5$; 512 unsafe everywhere	frozen criteria	verdict grid	§4.3
768: no certifiable window (softened from "never safe")	confirmatory (debiased)	verdict grid	§4.3
Ratio governs A_e; absolute size governs Floor_e	pre-registered contrast	iso-severity twin	§4.5
Residual mismatch route-uniform (kills N4, establishes A3)	pre-registered	residual probe	§4.5
Curve shape = m/G read	post-hoc consistency	renormalization	§4.5
Floor = RoPE + Resid split	confirmatory	PI intervention	§4.6
Account predicts held-out routes (~0.07–0.09 RMSE)	pre-registered gates	held-out validation	§4.7
Floor law 2-for-2 on unseen floors	gate + post-hoc addendum	leave-one-out	§4.7
Exact angular link is the best form	post-hoc (refit)	three-form refit	§4.7
768 mid-σ dip = negative interaction I_e	interpretation; direct probe pending	demote–re-promote	§4.7
~14% wall-clock ceiling	projected until A/B	fixed-step A/B	§5.2

Table 5: The measured safety map — *per-example* object (the worst case, what a batch-1 user sees). Floor_e = debiased endpoint floor where the debiased instrument ran; † marks routes measured only by the pre-debiasing instrument (the finite-draw bias is strictly positive, so their ≈ 0 floors and in-band verdicts are conservative reads; §4.2). σ^* = noise level above which the paired debiased excess is non-inferior at instrument resolution (Eq. 3); token ratio = per-forward cost of the demoted grid.

route	edge ratio	target tokens	Floor_e	σ^*	verdict
1024→896	0.875	~3,000	0.02–0.04	≈ 0.5	**safe**, $\sigma \in (0.5, 0.94]$
1280→1024†	0.80	~4,100	≈ 0	$\in (0.625, 0.875)$	**safe**, $\sigma \gtrsim 0.75$
1280→1120†	0.875	~4,800	≈ 0	≈ 0.5	safe (synthetic grid)
896→768†	0.857	~2,160	0.06–0.12	—	no certifiable window
1024→768	0.75	~2,160	0.06–0.09	—	no certifiable window
any →512	≤ 0.57	~1,000	≈ 0.30	—	certifiably unsafe, all σ

5 The answer, in deployable form

5.1 The (route, σ) safety map

Table 5 is the answer to the title question at our operating point: what replaces the spectral rule is not a corrected formula but a measured map at a stated resolution. Safety is a per-route boundary $\sigma^*(\text{route})$, jointly determined by ratio (through A_e) and absolute target size (through Floor_e); neither alone predicts the ordering. For the corpus safe route the fine print is: every bin in $(0.5, 0.94]$ has a debiased mean within ±0.03 of zero, formal certification at the strict 0.02 margin is achieved at $\sigma = 0.688$ (elsewhere bin-level ε^* exceeds it at the current N, D), and the σ=1 endpoint itself carries a small real gap (+0.042 ± 0.011) — the safe window is the interior.

The map depends on the aggregation object. Per §3.1, the aggregation operator is part of the estimand, and the two objects genuinely disagree at high σ: pooled-arm probes (pool size 4) collapse the 1024→896 excess to ≈ 0 at every bin $\sigma \geq 0.625$ *and* the 1024→768 excess to ≈ 0 at $\sigma \geq 0.875$ — the batch-aggregate object SGD follows is more forgiving than the per-example worst case, and the small per-image residual behaves as draw noise that cancels across images rather than as a shared bias. We publish the per-example map as the verdict (it bounds every batch size from below; pooled arms do not yet have self-floors), and a debiased verdict grid at the shipped trainer's actual batch×accumulation operating point is designed **[pending]**. A redundancy-stratified analysis finds no per-image demotion-targeting lever (null trend across strata), consistent with the floor being an image-generic network property.

5.2 σ-conditional training on the measured-safe route

We wire the one measured-safe corpus route into the LoRA trainer as an opt-in path, exactly as measured — no extrapolation beyond the map:

σ-first draw. The trainer draws each batch's σ *before* fetching latents, from the unchanged training density (the σ marginal is untouched — demotion conditions on σ, never the reverse). When every sample in the batch is above the gate ($\sigma > 0.5$), the native 1024-tier latent is swapped for a cached demoted-grid sibling produced by the probe's own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream — noise, target, masking, loss — derives from the swapped latent. Off-route images always train native, as does validation. The small real gap at the exact σ=1 endpoint (§4.4) does not bite here: it is a boundary point of measure zero for the continuous training σ-density, and the last interior bin ($\sigma \in (0.875, 1)$) reads ≈ 0 debiased.

Banded rotary alignment, σ-gated. Motivated by §4.6, a frequency-selective alignment (YaRN-style bands: full positional-interpolation stretch for rotary bands with few rotations across the demoted extent, native spacing for high-frequency bands, ramp between, with the band thresholds scheduled by a sigmoid gate in σ following Zhao et al., 2026) is applied on demoted forwards only. Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs — it erases the static variant's low-σ liability ($+0.064 \to +0.033 \pm 0.025$) while preserving its high-σ gains (best and most stable arm at $\sigma \geq 0.59$ and the endpoint, paired -0.05 vs plain demotion). It is a refinement candidate, not a gate-widener: the gate stays $\sigma > 0.5$.

Cost accounting. With ~50% of draws above the gate and the demoted forward at ~0.72× tokens, the fixed-step epoch cost is $\approx 0.5 \cdot 0.72 + 0.5 = 0.86$ — a *projected ceiling* of ~14% wall-clock on our corpus (96% of records are on the safe route), scaling with high-tier content; the realized saving (including cache-load overheads) is measured only by the pending A/B below. The pitch is wall-clock at fixed steps, never "more steps in the same time."

Weight-space sanity. Under bit-exact deterministic training (paired runs; without determinism, twin runs put a chaos floor of 0.41 on checkpoint-displacement cosine, which would swamp the read), the rotary refinement adds no displacement over the demotion arm it refines (base↔refined 0.319 vs base↔demoted 0.320), with its footprint concentrated in the low-signal early/mid blocks — consistent with §4.6's depth profile.

What this section does not claim. The end-to-end acceptance gate — a fixed-step A/B at training scale scoring generation-quality non-inferiority (CMMD; Jayasumana et al., 2024) against the realized wall-clock saving — is designed and pre-committed (a regression closes the line) but not yet run; all claims above are gradient-level and weight-space-level. We flag this deliberately: the gap between gradient-level equivalence at N=40 and thousands of optimizer steps is exactly the kind of gap this paper is about.

6 Limitations

One model family, one operating point. All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint first is the designated bound on how far the map generalizes. The floor's dependence on the fine-tuning operating point through J is untested (the designated probe: fine-tune at the 1280 tier, re-probe 1280→1024). Relatedly, the probe pool's relationship to the adapter's own fine-tune set was never a controlled axis (2 of 40 verdict stems are in it); a controlled-adapter 2×2 factorial — two LoRAs trained on opposite style clusters under frozen stem-level membership manifests, probed on both clusters — measured this dependence rather than conceding it: the map's shape replicates on both checkpoints and the adapter×probe-style interaction is below instrument resolution, while the absolute redraw-floor level is checkpoint-dependent (Appendix D).

Gradient-level, not end-task. The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed only by the pending A/B above.

Account resolution and domain. The account predicts held-out routes at ~0.07–0.09 RMSE — never within ε^* (§4.7); its $A(\text{ratio})$ governor rests on two distinct ratio values; the reduction's domain excludes the 768 mid-σ window pending the direct interaction probe; and the band-counting sketch for $\text{RoPE}_e/\text{Resid}_e$ is calibrated, not derived.

Instrument resolution. "Safe" is one-sided non-inferiority at the measured certification resolution (§3.1): $\varepsilon^* = 0.03$–0.08 per bin at the verdict grid's (N, D), ≈ 0.01–0.02 at the endpoint sweeps. True gaps below ε^* are undetectable by construction, the σ^* boundaries are localized only to a bin, and every "no certifiable window" claim is (N, D)-relative — a larger campaign could certify (or refute) the 768 high-σ window whose debiased means already sit near zero.

Debiased coverage. The debiasing campaign covered the corpus verdict grid: the endpoint draw sweeps, the 8-bin map, x-zero, and the α-sweep. The 1280-tier iso-severity, 896-tier transfer, depth-split, and PI-intervention probes remain pre-debiasing reads, defended where used by the one-signed-bias argument and by paired same-grid contrasts (§4.2); pooled arms have no self-floors yet. The 1280-tier curves enter the held-out validation with this caveat recorded.

Scope. Adapter (LoRA) gradients on a frozen backbone; full fine-tuning may differ. Pixel-space demotion only (latent-space downsampling was not tested here; SwD found it inferior). The σ-gated banded-alignment parameters were taken from one shot at pre-registered values, not searched.

7 Conclusion

When does training on downscaled images yield the same gradients? Not when the noise has masked the frequencies the coarse grid loses — the spectral answer, even in its strongest tolerance-parameterized form, is a model of only one part of one branch of what demotion perturbs, and the measurements reject it at every point of structural divergence: the mildest route's crossover sits at 3.5× the equal-power prediction, no tolerance reproduces the map, and a 2×-downscaled latent is certifiably unsafe at every noise level, because the network function itself is grid-calibrated — partly through its positional coordinate system, partly through an irreducible dependence of the computation on token count. The measured answer has two conditions, matching the two terms of the account: the target grid must be large enough in *absolute token count* that its graph floor sits below certification resolution, and the noise level must exceed a per-route boundary where the ratio-governed data term dies. On our model that means 1024→896 at $\sigma > 0.5$ and 1280→1024 at $\sigma \gtrsim 0.75$ — and never a 2× downscale, at any noise level. The answer is predictive, not merely descriptive: the exact angular form of the account, with its two measured governors, predicts held-out routes at ~0.07–0.09 RMSE, and its floor law hit two floors it never saw. Getting there required treating the question as a metrology problem — naive finite-draw estimation manufactures exactly the token-count-ordered floor signature under test, half the intermediate route's raw floor was estimator bias, and our own headline claims were re-derived, and in one case softened, under the correction. On the one route our corpus makes safe, the map projects a ceiling of ~14% wall-clock at fixed steps, pending the end-to-end A/B. We offer the account and its instrument — per-σ-bin gradient agreement with a redraw floor, a re-encoding control, per-arm self-floors, and a stated certification resolution, criteria frozen before data — as the cheap, general test that any future "train part of the time at lower resolution" proposal should pass before it is believed; without the debiasing, that test manufactures the very effect it looks for.

Reproducibility statement

The probe scripts (`run_sigma_probe.py` — including the `--self_floor` and `--draw_sweep` debiasing modes — `run_prior_distance.py`, `rapsd.py`, `compare_ckpt_dw.py`), the held-out validation and refit analyses (`e5_holdout.py`, `e5_refit.py`), the trainer wiring (`--sigma_lowres`), and the frozen pre-registration documents (including the debiasing campaign's decision rules) are in the public repository at `https://github.com/sorryhyun/anima_lora` (project line `project/sigma_lowres/`). The debiased verdict runs' complete per-image records (`result.json`

+ `per_image.jsonl`) ship in-repo under `project/sigma_lowres/paper_bench/runs/`; the earlier raw-instrument run archives are excluded by the repository's results ignore rule and will be published as a companion tarball **[pending]** — until then, raw-run numbers are reproducible from the scripts but their original records are not themselves public. Paired training runs use a `--deterministic` mode (deterministic attention backward included) verified bit-identical across twin runs; finite-draw cosines are kernel-path-sensitive, so all cosine pairings are computed within a single run (Appendix C).

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A Derivation of the branch decomposition (Eq. 4)

Fix a query (image, caption), a σ, and an arm a. Per draw ϵ, the adapter gradient of the $\frac{1}{2}$-scaled loss is the product $g_a(\epsilon) = J_a(\epsilon)^\top r_a(\epsilon)$ of §3.2. Split each factor into its noise mean and fluctuation, $J_a = \bar{J}_a + \tilde{J}_a$ and $r_a = \bar{r}_a + \tilde{r}_a$ with $\mathbb{E}_\epsilon[\tilde{J}_a] = 0$ and $\mathbb{E}_\epsilon[\tilde{r}_a] = 0$; the population mean gradient of arm a is then

$$\bar{g}_a \;=\; \mathbb{E}_\epsilon\big[J_a^\top r_a\big] \;=\; \bar{J}_a^\top \bar{r}_a \;+\; \mathbb{E}_\epsilon\big[\tilde{J}_a^\top \tilde{r}_a\big]. \tag{10}$$

Write the demoted arm's mean factors as native plus a perturbation, $\bar{J}_e = \bar{J} + \Delta J$ and $\bar{r}_e = \bar{r} + \Delta\bar{r}$ (bars without an arm subscript denote the native arm). Subtracting Eq. 10 across arms and expanding the product,

$$\delta g \;=\; \bar{g}_e - \bar{g}_{\text{nat}} \;=\; \underbrace{\bar{J}^\top \Delta\bar{r}}_{B_e} \;+\; \underbrace{\Delta J^\top \,\bar{r}}_{C_e} \;+\; \underbrace{\Delta J^\top \Delta\bar{r} \;+\; \mathbb{E}_\epsilon\big[\tilde{J}_e^\top \tilde{r}_e\big] - \mathbb{E}_\epsilon\big[\tilde{J}_{\text{nat}}^\top \tilde{r}_{\text{nat}}\big]}_{R_e}\,. \tag{11}$$

This is an exact identity — no linearization is involved. "First-order" in the main text refers only to the content of R_e: it collects every term that is a product of two perturbations ($\Delta J^\top \Delta\bar{r}$) or a difference of noise-covariance terms (the last two). Eq. 4 is Eq. 11 with the mean-Jacobian bar suppressed in the notation.

One remark delimits the identity's literal scope, expanding the qualification stated in §3.2. When the two arms share a grid (as for the re-encoding control), every operation above is literal. Across grids, J_{nat} and J_e act on different token counts, so the differences ΔJ and $\Delta\bar{r}$ require a fixed identification of the two output spaces (e.g. spectral zero-padding of the coarse grid onto the fine grid's bands); any such identification changes the split between B_e, C_e, and R_e but not their sum δg, which lives in the shared adapter-parameter space regardless. This is why the main text defines the branches *operationally* — as the two independent perturbation directions in adapter-parameter space, isolated by the designated probes of §4.4 — and treats Eq. 4 as fixing what each branch collects rather than as a computable matrix formula.

B Derivation of the angular forms (Eqs. 5 and 6)

Both angular forms follow from Eq. 1 and the split of δg by the native direction; no property of demotion enters. Write $g = \bar{g}_{\text{nat}}$, $G = \|g\|$, $\hat{g} = g/G$, and decompose $\delta g = G\,\kappa_\parallel\,\hat{g} + P_\perp \delta g$ with $\kappa_\parallel = \hat{g}^\top \delta g / G$ and $\kappa_\perp = \|P_\perp \delta g\|/G$.

The exact form. The two components are orthogonal, so $\|g + \delta g\|^2 = G^2(1 + \kappa_\|)^2 + G^2\kappa_\perp^2$ and

$$\cos\left(g,\ g + \delta g\right) \;=\; \frac{\hat{g}^\top(g + \delta g)}{\|g + \delta g\|} \;=\; \frac{1 + \kappa_\|}{\sqrt{(1 + \kappa_\|)^2 + \kappa_\perp^2}} \;=\; \frac{1}{\sqrt{1 + \kappa_{\text{eff}}^2}}, \qquad \kappa_{\text{eff}} = \frac{\kappa_\perp}{1 + \kappa_\|}, \tag{12}$$

valid whenever $1 + \kappa_\| > 0$ (the perturbation does not reverse the gradient direction); $d_e = 1 - \cos$ is Eq. 5. Since $1 - \cos(\hat{g}_1, \hat{g}_2) = \frac{1}{2}\|\hat{g}_1 - \hat{g}_2\|^2$ for unit vectors, starting from either expression in Eq. 1 lands on the same result.

The quadratic limit. For small perturbations, $1 - (1 + \kappa_{\text{eff}}^2)^{-1/2} = \frac{1}{2}\kappa_{\text{eff}}^2 + O(\kappa_{\text{eff}}^4)$ and $\kappa_{\text{eff}}^2 = \kappa_\perp^2\left(1 - 2\kappa_\| + O(\kappa_\|^2)\right)$, so

$$d_e \;=\; \frac{\|P_\perp \delta g\|^2}{2G^2} \;+\; \underbrace{O(\kappa_\perp^2\kappa_\|) + O(\kappa_\perp^4)}_{\text{beyond-quadratic}}. \tag{13}$$

The four-term expansion. Substituting $\delta g = B_e + C_e + R_e$ (Eq. 4) into the leading term and expanding the square,

$$\|P_\perp \delta g\|^2 = \|P_\perp B_e\|^2 + \|P_\perp C_e\|^2 + 2\,\langle P_\perp B_e,\ P_\perp C_e\rangle + \Big(2\,\langle P_\perp (B_e{+}C_e),\ P_\perp R_e\rangle + \|P_\perp R_e\|^2\Big),$$

whose first three terms, divided by $2G^2$, are exactly the data share S_e, the graph share F_e, and the projected interaction I_e of Eq. 6. The remainder R'_e of Eq. 6 therefore has an exact content: the R_e-bearing terms above, the beyond-quadratic terms of Eq. 13, and the parallel-component correction — each suppressed by an extra power of perturbation size relative to the retained terms.

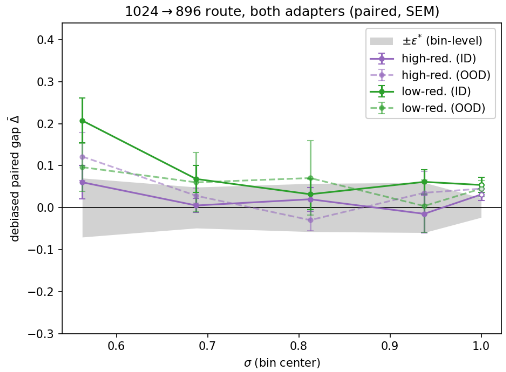
What is not derived. The loading $\|P_\perp \delta g\| = c_e\, m(\sigma)$ — that the orthogonal gradient mismatch is proportional to the residual mismatch with a σ-independent route gain — is an empirical ingredient (A3 at the amplitude level), fit and tested held-out in §4.7. The geometry above fixes only how a given δg is read into gap units.

C Instrument details

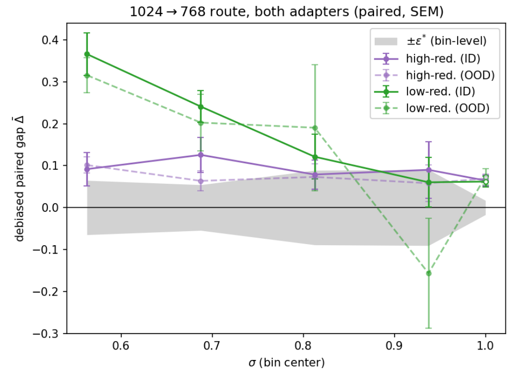
Estimator. Per image, arm, and σ-bin, adapter gradients are accumulated over D stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at D=8 are not comparable *across* bins (the floor drops where $\|g\|$ is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability ≈ 0 and is never used.

Controls. The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; in pre-debiasing runs its gap served as the validity band (± 0.04; observed $|\cdot| \le 0.054$ across all bins of the main runs, per-module-group ≤ 0.045 in the split runs), a role superseded by the paired debiased read of §4.2. The redraw floor prices finite-draw noise. Density-weighting the uniform bins by the trainer's σ-density reproduces earlier σ-marginalized measurements from the same instrument family (consistency check).

Debiasing (§3.5), implementation. Under `--self_floor`, every arm (re-encoding and each demote variant) runs a second independent draw set from a disjoint seed stream, giving per-arm self-cosines alongside the native floor; Eq. 9 is then computed per image and bin, and the map read is the paired excess $\Delta_i = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$ with non-finite values (negative self-cosines under the square root at low D) and $|\Delta_i| > 1.5$ trimmed (0–5 of 40 images per cell). Under `--draw_sweep`, draw prefixes are nested ($D = 64$ contains the $D \le 32$ sets; prefix sums, no extra forwards) and $\text{gap}(D) = \text{gap}_\infty + c/D$ is fit on route means with a bootstrap CI over images. Finite-draw cosines are kernel-path chaotic: twin runs sharing a warm compiler kernel cache agree to $|\Delta\cos| \le 0.015$, while a run compiled to a different kernel set lands up to ≈ 0.3 away at D=2 — so gap/floor/self-floor pairings are never compared across processes, a guarantee the single-run instrument provides by construction; a `--deterministic` mode (bit-exact across twin runs) covers the paired training A/Bs.



(a) 1024⟶896 (the certified route).



(b) 1024⟶768.

Figure 3: The Fig. 1c recipe repeated on two controlled adapters trained on opposite style clusters, one panel per route (N=48 stems per adapter, D=8/bin, high-σ window; same paired debiased estimator and trim; open markers the σ=1 endpoint bin; axes shared across panels). Unlike the main-text figures, color encodes the *adapter* (high- vs low-redundancy cluster); solid curves are in-distribution stems — the adapter's own style cluster (n=36: trained-on, held-out, and never-seen-artist cells) — and dashed curves are out-of-distribution stems from the opposite cluster (n=12). Gray band: bin-level $\pm\varepsilon^*$ per Eq. 2, median of the two runs' full-N SEMs for the panel's route. The map replicates: on the certified route all four adapter×distribution curves reach the band inside the window, on 768 all four sit above it until the top, and within each panel the dashed OOD curves track their solid ID counterparts — the adapter×probe-style interaction is not visible at this resolution.

Endpoint and x-zero modes. The σ=1 endpoint bin feeds input exactly ϵ (the input-mediated data term vanishes by construction). The x-zero mode zeroes x in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the $(1-\sigma)x$ term absent, low-σ bins are off-manifold and the σ=1 read is primary. In that regime the residual is $\approx -\hat{x}_{\text{prior}}$, so "graph-dominated" includes the model's grid-conditioned prior; the prior-distance probe (§4.4) subsequently dissociated the prior from the floor's route ordering.

Held-out validation and refit (§4.7), sources. The fit consumes the debiased paired curves of the 1024-tier verdict run, the 1280-tier curves (pre-debiasing paired reads; caveat recorded in §6), the route-uniform $m(\sigma)$ from the residual probe, and each run's own bin-mean $G(\sigma)$ (gap and G always from the same process, per the kernel-path rule). The shared-power fit scans $p \in [1, 2]$ jointly across fit routes with per-route weighted least squares; the exact-link fit is a per-route grid with profile-likelihood uncertainties on c_e. Oracle baselines are quadratic-in-σ fits on the held-out data itself (the noise ceiling for a 3-dof curve); the null baseline is gap = 0 on the null's committed region $\sigma > \sigma_{\text{eq}}$.

Spectral-null computation. Table 2 evaluates Eq. 8 at the demoted grids' Nyquist frequencies $f_{\text{cut}} = \frac{1}{2}\, e/e_{\text{nat}}$ on the N=40 mean radially-averaged latent power spectrum (64 radial bins; P at the 896/768/512 cuts = 0.025/0.029/0.065). The equal-power slice $\delta = (1 + P_\omega)/2$ recovers $\sigma_{\text{eq}} = \sqrt{P}/(1+\sqrt{P})$ exactly. Across $\delta \in [0.001, 0.5]$ the mildest–harshest spread never exceeds 0.13, and the 1024⟶896 versus 896⟶768 boundaries (cut frequencies 0.4375 vs 0.4286) never separate by more than 0.01. The parameter-free anchor δ_{reenc} of §4.3 is measured by the same estimator: RAPSD of the re-encode arm's latent error (fresh resize–encode minus cached latent, the gradient probe's own control chain) on the same probe set, interpolated at each route's cut frequency **[pending:** `reenc_noise_floor.py`**]**.

D Controlled-adapter replication of the verdict map

The one-adapter caveat of §6 pre-registered a controlled 2×2 factorial: two plain-LoRA checkpoints trained under the shipped probe adapter's frozen recipe on opposite style clusters — the top and bottom 12 artists by median latent redundancy, hereafter the high-redundancy (visually flat) and low-redundancy (heavily textured) clusters — with stem-level membership manifests frozen before training (124 training images each, identical seeds). Each adapter was probed on 48 stems spanning four membership cells (N=12 each): trained-on, held-out images of trained artists, never-seen artists of the same cluster, and never-seen artists of the opposite cluster, with the cross-cluster stems shared between the two runs so cross-adapter contrasts read paired per stem. Figure 3 repeats the verdict-map recipe per route, overlaying both adapters.

Three reads, stated at the bin-mean level (per-cell tables reproduce from the archived per-image rows). First, the (route, σ) shape replicates on both checkpoints (Fig. 3). Second, the factorial contrasts are null at instrument resolution: the probe-style main effect and the adapter×probe-style interaction — the pre-registered verdict quantity — are both bounded below the resolvable bin-mean effect (raw-gap paired interaction -0.022 ± 0.027 at 896, -0.015 ± 0.059 at 768), and the membership contrast (trained-on versus held-out) does not replicate in sign across the two adapters. The map is therefore adapter- and content-agnostic on this model at our resolution, supporting the calibrate-once-per-model reading of §5.1. Third, the one quantity that is *not* checkpoint-invariant is the absolute redraw-floor level: in-window floor cosines sit at ≈0.73 for the high-redundancy-cluster adapter versus ≈0.50 for the low-redundancy-cluster adapter, uniformly across all four membership cells including the opposite-cluster stems. The floor's level is a property of the checkpoint (its gradient stochasticity), while the safety map built on top of it is not.

E Historical (pre-debiasing) tables

The tables in this appendix are the raw-estimator record: retained for continuity and for the reads defended in §4.2 (one-signed bias; paired same-grid contrasts), never cited as debiased evidence.

Table 6: **Raw estimator** (historical record; superseded by Table 1): demotion gap by σ-bin, 1024-tier natives (N=40, bin-mean SEM ~0.02, re-encoding control $|\text{gap}_{\text{reenc}}| \le 0.054$ everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. Raw values understate mid-σ gaps (the floor attenuates the read) and overstate endpoint floors (single-estimate variance bias); the debiased map corrects both. The $\cos_{\text{floor}}$ and $\|g\|$ rows are the estimator context plotted in Fig. 1d.

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
gap_{896} (ratio 0.875)	.110	.162	.148	.137	**.048**	**.030**	**.030**	.053
gap_{768} (ratio 0.75)	.144	.216	.208	.223	.164	.163	.115	.063
gap_{512} (ratio 0.5)	.348	.410	.355	.469	.430	.391	.289	.296
$\cos_{\text{floor}}$	.84	.65	.51	.65	.70	.77	.80	.83
$\|g\|$ (native)	2.6	0.5	0.4	0.5	0.7	1.1	2.0	9.3

Table 7: Raw endpoint reads (D=16, N=40) beside their debiased draw-limit values (Table 3) — the debiasing's revision record: the 768 floor halves, the 896 floor surfaces.

route	raw endpoint (D=16)	endpoint $\widehat{\text{gap}}_\infty$ [95% CI]
native self-floor (cosine)	0.85	1.005 [0.994, 1.016]
re-encoding control	−0.038	−0.003 [−0.017, +0.008]
1024→896	−0.009	+0.019 [+0.010, +0.030]
1024→768	+0.127	+0.056 [+0.043, +0.071]
1024→512	+0.326	+0.304 [+0.197, +0.424]

Table 8: Iso-severity probe (**raw estimator**): gap by σ-bin on the 1280-tier cache (N=24, 4 draws/bin) against the corpus 1024→896 curve. The ratio-matched routes coincide within ~1.4 combined SEM at every bin despite a 1.6× difference in absolute target size; the same-native harsher-ratio route separates. *896's canonical 16-draw endpoint is −0.009 raw, +0.019 debiased (Table 3).

σ	0.125	0.375	0.625	0.875	1.0
re-encoding control	+.047	−.005	+.048	+.020	−.012
1280→1120 (ratio .875)	.129	.110	.054	−.012	−.007
1280→1024 (ratio .80)	.170	.178	.111	.002	.061
1024→896 (ratio .875)	.132	.076	.077	.049	.100*

Table 9: Ratio-transfer run (**raw estimator**): 896-tier natives (N=40), pre-registered bar "gap$_{768}$ within the re-encoding band at $\sigma \geq 0.5$" — **fails** (residual 0.06–0.12, ~2× the 1024→896 plateau; ~half of it is expected estimator bias, §4.2 — the separation is independently confirmed by the debiased same-grid floors, §4.4).

σ bin	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
896→768	.114	.209	.168	.143	.124	.056	.092	.061
896→512	.318	.372	.308	.340	.320	.280	.329	.217

Table 10: Route-uniformity of the mean prediction residual: cross-grid excess of $\bar{r} = \mathbb{E}_\epsilon[\hat{v} - (\epsilon - x)]$ over split-half floors (relative L_2; same-grid re-encoding control $\leq$ 0.02 everywhere). The three main routes coincide within $\sim \pm 0.02$ at every σ while their gradient floors span 0 to 0.3.

σ	0.125	0.375	0.625	0.875	1.0
1280→1024	.885	.825	.715	.465	.360
1024→896	.862	.808	.706	.459	.378
896→768	.860	.814	.715	.466	.393
768→512	.897	.872	.767	.508	.435

Table 11: Depth localization of the floor (**raw estimator**): mean per-block gap at σ=1, early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry ~3× the late-block gap. The apparent content share (ep − xz) is a late-block minority effect in raw units; the debiased draw-limit comparison finds ep = xz within CI at the whole-adapter level (Table 3).

route	ep early	ep late	xz early	xz late
1024→512	.357	.223	.351	.125
1024→768	.164	.085	.121	.033
1024→896	.023	.015	.044	.012

Table 12: Positional-interpolation intervention at the σ=1 endpoint (N=40, 16 draws; **raw** arm values — the paired Δ column is a same-grid contrast in which the finite-draw bias cancels to first order): exact phase-geometry alignment erases the mild-route floor and ~30% of the harsh-route floor.

route	plain (SEM)	PI-aligned (SEM)	paired Δ (SEM)	improved
1024→896 (control)	−0.021 (.041)	−0.040 (.040)	—	—
1024→768	+0.080 (.048)	−0.001 (.039)	+0.081 (.031)	78%
1024→512	+0.320 (.058)	+0.224 (.056)	+0.096 (.039)	70%